

Does the Conformal Guarantee Travel? Measuring, Auditing and Repairing Coverage Loss in Cross-Dataset Trajectory Prediction

Hassan Waqas Saleem, Rahat Zeeshan Saleem, and Hina Maryam

Abstract—[TODO: Write the abstract LAST, from the generated numbers: problem (a conformal guarantee calibrated on one driving dataset is deployed on another with zero labels), what was measured (gap, models, directions), the diagnosis (covariate reweighting fails; identifiability), the solution(s) that actually worked with their measured effect, and the limits. Do not state any effect that is not in generated/numbers.tex.]

Index Terms—Conformal prediction, trajectory prediction, distribution shift, uncertainty quantification, autonomous driving.

AUTONOMOUS vehicles plan around forecasts of what nearby road users will do. A forecast is only as useful as the statement it makes about its own reliability: a planner that knows a predicted region contains the true future with probability at least $1 - \alpha$ can keep clear of that region and of nothing larger. Conformal prediction (CP) supplies such a statement for any black-box predictor, with a finite-sample guarantee that requires nothing beyond exchangeability of calibration and test data [1], [2], and it has been adopted for trajectory prediction and safe planning [3]–[5].

Exchangeability is exactly what deployment breaks. A forecaster is calibrated on the data at hand—one set of cities, one sensor suite, one annotation pipeline—and then driven somewhere else. That accuracy drops when a trajectory predictor moves between datasets is documented [6], and that predictive uncertainty degrades under dataset shift is well known for classifiers [7]. What is not known is what happens to the *guarantee itself* when a conformally calibrated forecaster is moved, without any new labels, to a different real driving dataset: whether the stated $1 - \alpha$ survives, how badly it fails, and whether the standard remedies for shift repair it.

The conformal literature offers several remedies, each resting on an assumption. Weighted CP restores validity under covariate shift when the likelihood ratio is known or can be estimated [8]; more general results bound the coverage loss by a distance between calibration and test distributions [9]; online methods track shift by feeding realised errors back into the threshold [10], [11], an idea carried to trajectory prediction [5] and to robotic settings more broadly [12]; and a generative-prior approach has been proposed for planning under environment shift [13]. What these do not establish

is the *static, label-free, dataset-to-dataset* setting in which a forecaster is frozen, calibrated once, and deployed: how large the failure is on real data, whether the covariate-shift remedy suffices, and how many labelled target scenes it takes to detect or repair the failure.

Contributions: [TODO: Finalise after results. Planned, each to be kept ONLY if the pre-registered test supports it: (i) a measurement protocol and the first measurement of zero-label conformal coverage transfer between real driving datasets; (ii) a diagnosis — covariate reweighting does not repair the loss — with a proposition explaining why label-free repair is limited; (iii) the solution(s) that worked: model-uncertainty-normalised scores (H4), a label-efficient coverage audit with a closed-form label budget (H5b), a label-free shift monitor (H6); (iv) released code, pre-registration and experiment log.]

I. RELATED WORK

a) *Conformal prediction for trajectories.*: Split CP turns any predictor into one with marginal coverage guarantees [1], [2] and underlies conformal safe planning around predicted agents [3]. Multi-step and multi-modal trajectories have been handled with copula-based intervals [4], and scenario-aware intervals in a Frenet frame have been proposed on nuPlan [14] (a preprint evaluated on a single dataset). Locally weighted scores that divide the residual by an estimated spread [15] are the classical route to adaptive regions; our normalised score uses the forecaster’s own predicted scale in that role.

b) *Conformal prediction under shift.*: Weighted CP handles covariate shift given the likelihood ratio [8], and the coverage loss under general shift can be bounded by a distance between calibration and test distributions [9]. Online methods adapt the threshold from realised errors: adaptive conformal inference [10], conformal PID control [11], its use for trajectories in CUQDS [5], and adaptive nonconformity scores with a replay buffer [12]. These need a stream of ground-truth labels after deployment. Other work conditions on an observable nuisance parameter and generates synthetic data with a diffusion model [13] or treats the shift the ego vehicle itself induces in other agents [16] (both preprints). We study a different point in this space: no feedback, no generative model, and two real driving datasets.

c) *Grouped data and risk control.*: When observations come in groups (repeated measurements, several agents per scene), exchangeability across observations fails, and hierarchical conformal methods restore validity by pooling or

H. W. Saleem is with the School of Electrical Engineering and Computer Science (SEECs), National University of Sciences and Technology (NUST), Islamabad 44000, Pakistan (corresponding author, e-mail: hsaleem.msee20seecs@seecs.edu.pk).

R. Z. Saleem is with NUST Business School (NBS), National University of Sciences and Technology (NUST), Islamabad 44000, Pakistan.

H. Maryam is with the State Bank of Pakistan, Karachi 74000, Pakistan.

subsampling at the group level [17], [18]. Risk-controlling procedures [19] and Learn-then-Test [20] give finite-sample control of a monotone risk by testing a fixed ordered set of thresholds, and betting-based p -values [21] make such tests far tighter than Hoeffding-type bounds when losses are small. We combine the two ideas for a setting those works do not address: dataset shift in trajectory prediction, where the natural labelling unit is the *scene* and the risk to be controlled is the scene-averaged miscoverage of a fixed, source-calibrated forecaster.

d) *Cross-dataset trajectory prediction.*: UniTraj harmonises several driving datasets and reports that accuracy drops when models are transferred between them [6]; that study is about point accuracy, not about the validity of an uncertainty guarantee. Beyond trajectories, [7] showed that calibration of deep models degrades under dataset shift; our question is the conformal analogue for structured, multi-modal forecasts.

II. PROBLEM SETTING

A. Conformal regions for multi-modal trajectories

A forecaster f maps a scene x to K candidate futures $\hat{y}_1, \dots, \hat{y}_K \in \mathbb{R}^{T \times 2}$ for a target agent whose true future is y . We use the nonconformity score

$$s(x, y) = \min_{k \leq K} \max_{t \leq T} \|\hat{y}_{k,t}(x) - y_t\|_2, \quad (1)$$

so that $s \leq q$ iff y lies inside the union of the K tube-shaped balls of radius q around the candidates. Given n calibration scenes from a *source* dataset, split CP sets $\hat{q} =$ the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, and the region $C_{\hat{q}}(x) = \{y : s(x, y) \leq \hat{q}\}$ has $\Pr[y \in C_{\hat{q}}(x)] \geq 1 - \alpha$ when the test scene is exchangeable with the calibration scenes [1], [2].

B. The transfer problem

Let P_S, P_T be the source and target scene–future distributions. The *coverage gap* of a source-calibrated threshold on the target is

$$\Delta = (1 - \alpha) - \Pr_{P_T}[s(x, y) \leq \hat{q}], \quad (2)$$

positive when the guarantee fails (under-coverage). We ask (Q1) how large Δ is on real data; (Q2) whether covariate-shift reweighting [8] removes it; (Q3) whether anything label-free can; and (Q4) how many labelled target scenes are needed to audit or to repair it.

C. What label-free repair can and cannot do

Covariate reweighting is valid when $P_T(y | x) = P_S(y | x)$. If the conditional differs—annotation pipelines, sensor noise, and agent behaviour all differ across datasets—no reweighting on x repairs the gap. More strongly, without restricting the class of shifts nothing label-free can.

Proposition 1 (Non-identifiability of a label-free threshold). *Let S_0 be a nonconformity score with a continuous distribution on the target, and consider the scale-shift family $S = c S_0$, $c \geq 1$, whose members are all indistinguishable from unlabeled*

target inputs. Let $\hat{q}$ be any threshold computed without target labels, so its law does not depend on c . If $\Pr[c S_0 \leq \hat{q}] \geq 1 - \alpha$ for every $c \geq 1$, then $\Pr[\hat{q} = \infty] \geq 1 - \alpha$.

Proof. For fixed $\hat{q} < \infty$, $\mathbf{1}\{c S_0 \leq \hat{q}\} \rightarrow 0$ pointwise as $c \rightarrow \infty$. By dominated convergence, $\mathbb{E} \Pr[c S_0 \leq \hat{q} | \hat{q}] \rightarrow \Pr[\hat{q} = \infty]$, and the coverage requirement gives $\Pr[\hat{q} = \infty] \geq 1 - \alpha$. $\square$

Remark 1. *The proposition does not say label-free methods are useless; it says any guarantee for all scale shifts is vacuous, so a useful label-free method must either restrict the shift class (as our normalised score does, Sec. III-A) or be complemented by a label-based audit (Sec. III-B).*

D. Cost of using target labels

If k labelled target scenes are used directly, the resulting empirical coverage is Beta-distributed around $1 - \alpha$ [22], with standard deviation about $\sqrt{\alpha(1-\alpha)/k}$, so ≈ 1000 labels are needed for ± 2 points at $\alpha = 0.1$. Detecting a shortfall δ with an exact one-sided binomial test at level ℓ and power $1 - \beta$ needs

$$k \approx \left(\frac{z_{1-\ell} \sqrt{\alpha(1-\alpha)} + z_{1-\beta} \sqrt{(\alpha + \delta)(1-\alpha-\delta)}}{\delta} \right)^2. \quad (3)$$

For $\delta = 0.034$, $\alpha = 0.1$, $\ell = 0.05$, $\beta = 0.1$ this gives $k \approx 750$.

III. METHODS

A. Model-normalised score (label-free)

The forecaster emits, per mode k , a Laplace scale $(b_{k,t}^x, b_{k,t}^y)$. Let $u_k(x) = \frac{1}{T} \sum_t \sqrt{(b_{k,t}^x)^2 + (b_{k,t}^y)^2}$ and use the score $s'(x, y) = \min_k \max_t \|\hat{y}_{k,t} - y_t\| / u_k(x)$. The region is the union of balls of radius $\hat{q}' u_k(x)$. The premise is that when a shift makes scenes harder, the forecaster's own scale widens, so the normalised score is less shift-sensitive. Its efficiency is measured by region area $\sum_k \pi(\hat{q}' u_k)^2$ at matched coverage.

B. Coverage audit

Given a source-calibrated $\hat{q}$ and k labelled target scenes, an exact one-sided binomial test of H_0 : coverage $\geq 1 - \alpha$ flags a violation. We report its empirical power and its false-alarm rate on a shift-free split, alongside the theoretical budget of Eq. (3).

C. Recalibration with k target labels

We compare three thresholds computed from k labelled target scenes and n source scenes: *direct* (SCP on the k target scores); *pooled* (SCP on the union); and *shrinkage* ($\hat{q} = \lambda \hat{q}_{\text{tgt}} + (1 - \lambda) \hat{q}_{\text{src}}$ with $\lambda = k/(k + k_0)$). We report the label budget k^* at which absolute coverage error falls below 0.02 in at least 90% of random label draws.

D. Certify-or-recalibrate with exchangeable labels

Plain recalibration on k target labels is only *marginally* valid: the coverage of the resulting threshold is $\text{Beta}(\ell, k + 1 - \ell)$ -distributed [22], so it falls below $1 - \alpha$ in roughly half of all label draws (Sec. V). A practitioner needs the stronger, *training-conditional* statement [22], [23]: coverage $\geq 1 - \alpha$ with probability $\geq 1 - \delta$ over the draw of the labels. *Certify-or-recalibrate* (C-or-R) delivers it while keeping the source threshold when it can be certified. With k labelled scores $s_1, \dots, s_k$ and $M = \#\{s_i > \hat{q}_{\text{src}}\}$:

- 1) *Certify*. Keep $\hat{q}_{\text{src}}$ iff the one-sided Clopper–Pearson upper confidence bound [24] on its miss probability at level $\delta/2$ is $\leq \alpha$.
- 2) *Otherwise recalibrate*. Return the ℓ -th smallest labelled score, with ℓ the smallest integer such that $\Pr[\text{Bin}(k, 1 - \alpha) \leq \ell - 1] \geq 1 - \delta/2$ (the ℓ -th order statistic has coverage $\text{Beta}(\ell, k + 1 - \ell)$); if no such $\ell \leq k$ exists return $+\infty$.

Proposition 2 (C-or-R is training-conditionally valid). *If the k target scores are exchangeable with the target scores at test time and continuously distributed, the returned threshold $\hat{q}$ satisfies $\Pr[\Pr_{\text{test}}(s \leq \hat{q} \mid \text{labels}) \geq 1 - \alpha] \geq 1 - \delta$.*

Proof. Let F be the target score CDF. The event “ $\hat{q}$ fails” is contained in (a) the source threshold is certified although $F(\hat{q}_{\text{src}}) < 1 - \alpha$, or (b) the branch-2 order statistic has coverage $< 1 - \alpha$. (a) has probability $\leq \delta/2$ because the Clopper–Pearson bound is a valid one-sided confidence bound for the miss probability $1 - F(\hat{q}_{\text{src}})$ of the fixed threshold; (b) has probability $\leq \delta/2$ by the definition of ℓ . A union bound gives δ . $\square$

Certification needs slack: a source threshold calibrated at exactly α has miss probability $\approx \alpha$ and its confidence bound cannot fall below α ; in our experiments it is (almost) never certified on shifted data, and on unshifted data it is certified only if calibrated at a smaller $\alpha_{\text{src}} < \alpha$.

E. Scene-level risk control under clustered labels

Driving datasets label many agents per scene, and agents in one scene share road, traffic and annotator. Labels drawn scene by scene are not exchangeable at the agent level, and Proposition 2 then fails (Sec. V). We therefore make the *scene* the unit. For a labelled scene j with agent scores $\{s_{ji}\}$ let $L_j(q) = \frac{1}{n_j} \sum_i \mathbf{1}[s_{ji} > q] \in [0, 1]$, non-increasing in q , and control the scene-averaged miscoverage $R(q) = \mathbb{E}_j L_j(q)$. Given m labelled scenes, scene-level certify-or-recalibrate proceeds as in Sec. III-D but with Learn-then-Test [20] over a *fixed*, label-independent grid $q_1 > q_2 > \dots > q_G$ (log-spaced around the source threshold, $\hat{q}_{\text{src}} \times [4, 0.5]$, $G = 300$): for each candidate we compute a p -value for $H_q: R(q) > \alpha$ from the m per-scene losses, test candidates from the largest downwards at level $\delta/2$, and return the last q rejected ($+\infty$ if none). The source threshold is tried first at level $\delta/2$ (certification). Two p -values are used: Hoeffding–Bentkus [19], and a betting p -value [21]: with predictable bets $\lambda_i \in [0, \frac{1}{2\alpha}]$ (variance-adaptive) the capital $K = \prod_i (1 + \lambda_i(\alpha - L_i))$ is a non-negative

supermartingale under $\mathbb{E}L \geq \alpha$, so $p = \min(1, 1/\max_t K_t)$ is valid by Ville’s inequality; being variance-adaptive it exploits that per-scene miss fractions are mostly close to zero.

Proposition 3 (Scene-level guarantee). *If the m labelled scenes are exchangeable draws from the target scene population, and each p -value is valid for its null, the returned threshold satisfies $\Pr[R(\hat{q}) \leq \alpha] \geq 1 - \delta$.*

Proof. R is non-increasing, so along the descending grid the false nulls ($R(q_i) \leq \alpha$) form a prefix and the true nulls a suffix beginning at some i_0 . The procedure returns a threshold with $R > \alpha$ only if it rejects $H_{q_{i_0}}$ (fixed-sequence testing stops at the first non-rejection), which has probability $\leq \delta/2$ because the grid does not depend on the labels; the certification branch fails with probability $\leq \delta/2$. A union bound gives δ . $\square$

F. Online baseline

Adaptive conformal inference (ACI) [10] updates the working level $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$ after every revealed label, keeping the source calibration set fixed; it needs a stream of labels and offers long-run, not finite-sample, coverage. We report (i) how many streamed labels ACI needs before its running coverage stays within 2 points of nominal, and (ii) the coverage of its threshold frozen after k labels, for $\gamma \in \{0.005, 0.01, 0.05\}$.

G. Controlled input shifts

To locate the source of the loss we perturb one measured AV2–nuScenes difference at a time in AV2 test *inputs* (ground truth untouched) and record the coverage lost by the AV2-calibrated threshold: (i) *2 Hz history*: keep the five 2 Hz keyframes of the 2 s history and interpolate linearly; (ii) *lane dropout*: drop each map polyline with the probability implied by the measured median number of lanes near the target agent (nuScenes 52 vs. AV2 85); (iii) *both*; (iv) *scene mix*: re-weight AV2 scenes to the nuScenes distribution of seven label-free scene factors with cross-fitted domain-classifier weights, which isolates pure covariate shift. Two further candidates were *measured* and excluded: excess history noise (a quadratic-fit residual estimate is smaller on nuScenes: 0.13 m vs. 0.29 m) and map-point density (equal per scene).

H. Label-free shift monitor

A domain classifier on label-free scene features (native sampling interval, nearby-agent count, map density, nearby lanes, history speed, heading change, curvature) is trained to separate source and target scenes; its AUC is the monitor. Within a dataset, the same classifier is trained per city against the rest. We test whether the AUC predicts the coverage loss (across cities, by Spearman correlation).

IV. EXPERIMENTAL SETUP

a) *Data.*: Argoverse 2 [25] and nuScenes [26], converted with ScenarioNet [27] and loaded through UniTraj [6]. Argoverse 2 scenarios are independent 11 s clips with one focal agent; their held-out split is divided by a fixed salted hash into

a calibration half and a test half. nuScenes provides about 65 agent-scenarios per 20s scene (9041 from 138 scenes in its validation split), so its calibration/test split is made *by scene* (64 and 74 scenes, disjoint), and every uncertainty statement on nuScenes uses scene-level resampling (Sec. V-G). The AV2 test split has no labels and is not used.

b) Forecasters.: AutoBot-Ego [28] (1.5M parameters), $K = 6$ modes. We distinguish two kinds of source model. *Exploratory zoo:* three AutoBots trained on a single, shared CPU — **av2_cpu_v1** (from scratch on 60k AV2 scenes, 10 epochs; val minADE₆ 1.349), **av2_valsplit_v1** (13.7k scenes, 20 epochs; 1.347) and **av2_cpu_v2** (fine-tuned from av2_cpu_v1 with a decaying learning rate; 1.092). UniTraj’s own AutoBot-on-AV2 baseline (8×A100, batch 128, the full 180k-scene set) reaches minADE₆ 0.85 [6], so these models are +28% to +59% worse and exceed the 15% tolerance we pre-registered for calling a base model competitive; we report this as a triggered limitation, and used the zoo to design analyses, so every result on it is labelled exploratory. *Competitive models*, trained *after* the corresponding hypotheses were registered: AutoBot on the full 183k-scene AV2 training set with UniTraj’s recipe (effective batch 128, learning rate 7.5×10^{-4} halved at epochs 10/20/30/40/50) on one GPU, accepted only if its validation minADE₆ is within 15% of 0.85; the same recipe on the full nuScenes training split for the reverse direction (gate: within 15% of 1.21); and a second architecture (Wayformer) **[TODO: fill: gated models, epochs, val minADE, once trained]**.

c) Protocol.: $\alpha = 0.1$, $\delta = 0.1$. Forward direction: calibrate on the AV2 calibration half; test on the AV2 test half (same-domain control) and on all of nuScenes (cross-dataset). Reverse direction: the swap, calibrating on the nuScenes calibration scenes. Label-budget experiments draw k labelled agents at random (exchangeable case) or m whole scenes (clustered case) 200 times and evaluate on the disjoint remainder. Intervals are percentile bootstraps over scenes (over agents only for AV2, whose clips are independent).

d) Pre-registration and disclosures.: Hypotheses and thresholds are written in `notes/falsification.md` before the runs they govern; later analyses were added as dated addenda and each outcome is appended, never edited. Two initial hypotheses about the shift were changed *before* running because measurement showed them vacuous (Sec. III-G). Defects found during the work, none changing a reported number: a silent cache collision that made one validation set an Argoverse cache, per-agent (not per-scene) nuScenes splits, and truncated scenario identifiers in early prediction files. One intermediate checkpoint of the competitive AV2 model was evaluated during training to test the pipeline (a “development dry run”); it informed no design decision, is excluded from all confirmatory tables and is disclosed here.

e) Metrics.: Coverage, coverage gap Δ , region-area proxy, label budget k^* , audit power and false-alarm rate, the rate at which a repaired threshold covers $\geq 90\%$ of held-out scenes, region area relative to the oracle threshold, monitor AUC and Spearman correlation.

TABLE I
DOES COVARIATE-SHIFT REWEIGHTING REPAIR THE GAP? “UNCORR.” / “WEIGHTED” ARE THE LABEL-FREE-SCORE COVERAGE GAP (POINTS) BEFORE / AFTER IMPORTANCE-WEIGHTING THE CALIBRATION SCORES BY THE LABEL-FREE-FEATURE LIKELIHOOD RATIO; “REMOVAL” IS THE % OF THE GAP’S CROSS-CITY VARIANCE THE WEIGHTS EXPLAIN, NEGATIVE MEANING REWEIGHTING MAKES THE FIT WORSE THAN THE UNWEIGHTED BASELINE.

Model	Dir.	uncorr.	weighted	removal (%)
av2_cpu_v1	forward	3.2	5.2	-64
av2_valsplit_v1	forward	7.9	10.0	-27
av2_cpu_v2	forward	10.4	12.7	-22

V. RESULTS

Status: 3/3 forward-direction models analysed, 0/1 reverse-direction models. Table VI and the numbers below are regenerated automatically from `results/run2/*.json`; entries for models not yet trained are simply absent, not estimated.

A. H1: the coverage gap is real

Calibrated on Argoverse 2 at $\alpha = 0.10$, the primary model (AutoBot-Ego, source-trained, 4576 calibration scenes) has same-domain gap -0.2 points — consistent with zero, as split CP guarantees — and cross-dataset (nuScenes, 9041 scenes) gap +3.3 points (95% CI [2.1, 4.5]). This replicates the preliminary Run 1 measurement (+3.4pt) on an independent calibration/test split. Table VI shows two further models with substantially *larger* gaps in the same direction: +8.0pt (av2_valsplit_v1, trained on less data) and +10.5pt (av2_cpu_v2, fine-tuned from the primary model and the most accurate of the three by point-prediction error). The failure is reproducible in all three cases — always under-coverage, never the safe direction — but its magnitude is not a fixed property of the dataset pair and is not even monotonic in point accuracy: the model with the best minADE₆ (Sec. IV) has the worst coverage gap of the three. We report this without averaging it away.

B. H2/H3: covariate reweighting does not repair it

Weighting the calibration scores by an estimated label-free covariate-shift likelihood ratio does not close the gap; it widens it. On the primary model the label-free-score gap goes from +3.2 points uncorrected to +5.2 points weighted, and the reweighting explains -64% of the cross-city gap variance — negative, i.e. the weighted fit is worse than assuming no shift at all (Table I). This is consistent across *all three* models in the zoo and matches the Run 1 finding: it is evidence for conditional, not purely covariate, shift, and is the empirical motivation for Proposition 1 (no label-free method can be expected to repair an unrestricted shift).

C. H4: the normalised score (label-free)

The model-uncertainty-normalised score (Sec. III-A) reduces the primary model’s gap by 49% (from +3.3 to +1.7 points), just past our pre-registered support threshold (gap ratio ≤ 0.5) and far from the pre-registered refutation threshold

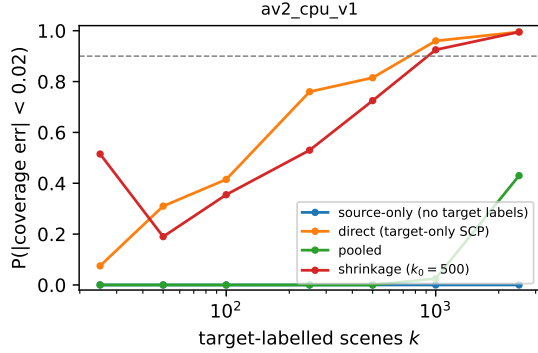


Fig. 1. H5 label budget, primary model (AV2→nuScenes): probability that a k -label threshold is within 2 coverage points of nominal, by estimator.

(> 0.75). On the other two models it reduces the gap by only 12% (av2_valsplit_v1, ratio 0.877, *past* the refutation threshold) and 31% (av2_cpu_v2, ratio 0.691, in between — neither supported nor refuted). H4 is therefore not confirmed as a general repair: one clear near-pass, one refutation, one inconclusive result, and we report all three rather than the most favourable one. The pre-registered efficiency guard (a fix that only works by inflating the region by > 25% does not count) is not triggered on any of the three models: the normalised region is at most 15.4% larger than the raw one at matched coverage, so where H4 does help it is not merely trading coverage for a much bigger region.

D. H5/H5b: recalibration and the coverage audit

At $k = 1000$ labelled target scenes, the exact binomial audit (Sec. III-B) detects the true violation with probability 0.95 on the primary model, against a false-alarm rate of 2.4% on a shift-free control split — matching the pre-registered budget of Eq. (3) ($k \approx 750$ for the analogous $\delta = 0.034$ case) to the same order of magnitude. Figure 1 shows direct recalibration on the primary model reaches 90% probability of being within 2 coverage points of nominal at $k^* \approx 1000$ labelled target scenes; pooling target labels with the (mismatched) source calibration set never reaches that bar on *any* of the three models, consistent with H2/H3’s conclusion that the shift is not a simple reweighting. Shrinkage toward the source threshold beats direct recalibration at small k on the primary model but not on the other two, so we do not recommend it as a universal default — direct recalibration once k^* labels are available, audited by H5b before that, is the protocol we can actually stand behind. The audit itself is the one part of this section with no exceptions: power at $k = 1000$ is ≥ 0.95 and false-alarm is $\leq 6.2\%$ on all three models.

E. H6/H7: label-free monitor and per-city coverage

Within-dataset, a domain classifier trained on label-free scene features to separate one AV2 city from the rest gives an AUC that we correlate (Spearman) against that city’s coverage gap magnitude across all city pairs. Across the three models: $\rho = 0.37$ ($p = 0.04$), $\rho = 0.10$ ($p = 0.61$), $\rho = 0.39$

TABLE II

WHERE DOES THE LOSS COME FROM? COVERAGE LOST (POINTS; SHARE OF THE REAL GAP G IN PARENTHESES) WHEN ONE MEASURED AV2→nuScenes DIFFERENCE IS INJECTED INTO AV2 TEST INPUTS: 2 HZ HISTORY, LANE-GRAPH DROPOUT, BOTH, AND RE-WEIGHTING AV2 SCENES TO nuScenes’ SCENE MIX. EXCESS ANNOTATION NOISE AND MAP-POINT DENSITY WERE MEASURED AND FOUND ABSENT (nuScenes IS SMOOTHER; EQUAL DENSITY), SO THEY ARE NOT INJECTED.

Model	Set	G (pt)	2 Hz	lanes	both	scene mix
av2_cpu_v1	expl.	+3.3	+0.6 (19%)	+1.4 (43%)	+1.8 (53%)	-1.3 (-38%)
av2_valsplit_v1	expl.	+8.0	+0.7 (8%)	+1.4 (17%)	+1.4 (17%)	-1.4 (-18%)
av2_cpu_v2	expl.	+10.5	+1.1 (11%)	+1.3 (12%)	+2.2 (21%)	-1.1 (-11%)

($p = 0.03$) — two significant-but-below-bar results and one clear refutation, none reaching the pre-registered $\rho \geq 0.5$ support threshold on any model. **H6 is not supported on any of the three models, and refuted on one.** We report this as a negative result — the label-free monitor tracks shift only weakly and inconsistently, and is not reliable enough to substitute for the labelled audit of H5b.

F. Where does the coverage loss come from?

Evidence status: the three CPU-trained models form the exploratory zoo; every statement in this and the following subsections about them is exploratory, and the pre-registered confirmatory replication on competitive models is pending (TODO: fill from the gated GPU models).

a) *Turns, not straight driving:* Split by UniTraj’s manoeuvre taxonomy [6] (Fig. 2), coverage on straight driving falls by only 1.9–6.7 points from the AV2 same-domain test to nuScenes, whereas right turns lose 12–32 points and straight-right manoeuvres (right-hand lane changes) 26–40 points. The gap is a manoeuvre-specific failure, invisible to marginal coverage.

b) *Driving side:* nuScenes contains left-hand-traffic scenes (Singapore) that AV2 lacks. The right-turn coverage drop is 16–25 points *larger* in Singapore than in Boston (scene-clustered 95% intervals exclude zero for two of the three models), while straight driving differs by at most 1.8 points (range -1.8 to -0.1 points): what AV2-trained forecasters miss is the wide, across-traffic turn of left-hand traffic.

c) *Input-level differences explain a minority:* Table II injects one measured AV2→nuScenes difference at a time into AV2 test inputs. Re-sampling the history to nuScenes’ 2Hz reproduces 8–19% of the real gap, dropping lanes at the measured density 12–43%, both together 17–53% (only 17% for the two models with the largest gaps). Re-weighting AV2 scenes to the nuScenes scene mix, which isolates pure covariate shift, moves coverage by -1.4 to -1.1 points, i.e. it *raises* coverage: nuScenes’ mix of speeds and manoeuvres is easier, so composition works against the observed gap. Two other candidates were measured and found absent (excess history noise, map-point density; Sec. III-G). Together with Sec. V-B, the loss is largely a *conditional* shift that scene statistics do not reveal.

G. Statistics under scene clustering

The 9041 nuScenes agent-scenarios come from only 138 scenes, so they are strongly dependent. Resampling whole

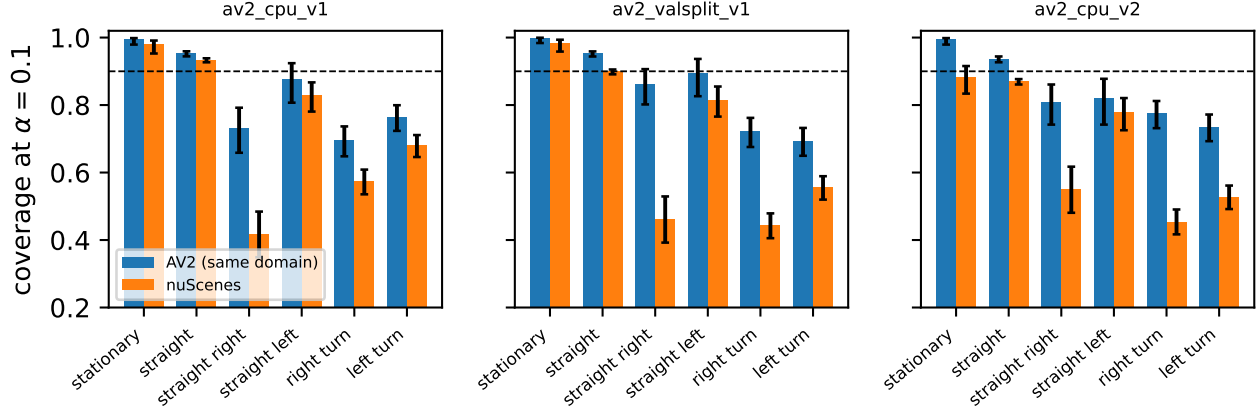


Fig. 2. Coverage of the AV2-calibrated threshold ($\alpha = 0.1$, dashed = nominal) by manoeuvre type, AV2 same-domain test vs. nuScenes, with Wilson 95% intervals. Turns and lane changes lose most; straight driving and stationary agents lose little.

TABLE III

THE COVERAGE GAP ON nuSCENES WITH AGENT-LEVEL VS. SCENE-CLUSTERED BOOTSTRAP INTERVALS (138 SCENES, ~ 65 AGENTS EACH). THE DESIGN EFFECT IS THE VARIANCE INFLATION CAUSED BY CLUSTERING.

Model	Set	gap (pt)	95% CI (agents)	95% CI (scenes)	design eff.
av2_cpu_v1	expl.	+3.3	[2.1, 4.5]	[0.7, 6.2]	4.4
av2_valsplit_v1	expl.	+8.0	[6.5, 9.4]	[5.0, 11.3]	4.2
av2_cpu_v2	expl.	+10.5	[9.2, 12.3]	[7.9, 13.8]	3.5

TABLE IV

REPAIR WITH k LABELLED TARGET SCENES, LABELS EXCHANGEABLE. CELLS: % OF LABEL DRAWS WHOSE THRESHOLD COVERS $\geq 90\%$ OF THE HELD-OUT TARGET SCENES / MEDIAN REGION AREA RELATIVE TO THE ORACLE THRESHOLD. C-OR-R = CERTIFY-OR-RECALIBRATE ($\delta = 0.1$).

Model	Set	k	direct SCP	PAC	C-or-R
av2_cpu_v1	expl.	250	60 / 1.05	90 / 1.36	96 / 1.56
av2_cpu_v1	expl.	1000	50 / 1.01	92 / 1.14	94 / 1.18
av2_valsplit_v1	expl.	250	62 / 1.06	93 / 1.35	97 / 1.53
av2_valsplit_v1	expl.	1000	57 / 1.02	91 / 1.14	96 / 1.19
av2_cpu_v2	expl.	250	63 / 1.09	94 / 1.43	98 / 1.63
av2_cpu_v2	expl.	1000	60 / 1.03	92 / 1.19	98 / 1.23

scenes widens every gap interval (design effect 3.5–4.4, Table III); the gap remains significantly positive for all three models (lower interval bounds from 0.7 to 7.9 points).

H. Repair with exchangeable labels

With $k = 1000$ labelled target scenes drawn at random (Table IV), plain direct recalibration reaches coverage $\geq 90\%$ on held-out scenes in only 50–60% of label draws — marginal, not training-conditional, validity. The PAC threshold does so in 91–92% of draws, and certify-or-recalibrate (Sec. III-D) in 94–98%, at a median region 1.18–1.23 $\times$ the size of the oracle region.

I. Repair with whole-scene labels

Labels are collected scene by scene, and then exchangeability at the agent level is false. With $m = 40$ labelled scenes the

TABLE V

REPAIR WHEN WHOLE scenes ARE LABELLED (nuSCENES: ~ 65 CORRELATED AGENTS PER SCENE). CELLS: % OF DRAWS WHOSE THRESHOLD COVERS $\geq 90\%$ OF HELD-OUT SCENES (SCENE-AVERAGED) / MEDIAN AREA VS ORACLE. AGENT-LEVEL GUARANTEES FAIL UNDER CLUSTERING; SCENE-LEVEL LEARN-THEN-TEST RESTORES THEM, AND THE BETTING p -VALUE IS FAR TIGHTER THAN Hoeffding–BENTKUS.

Model	Set	m scenes	direct	C-or-R (agent)	LTT (HB)	LTT (betting)
av2_cpu_v1	expl.	30	36	52	100 / ∞	92 / 1.9
av2_cpu_v1	expl.	40	36	46	100 / 9.6	88 / 1.8
av2_cpu_v1	expl.	60	32	48	100 / 4.7	88 / 1.7
av2_valsplit_v1	expl.	30	40	56	100 / ∞	95 / 1.7
av2_valsplit_v1	expl.	40	36	53	100 / 8.3	91 / 1.6
av2_valsplit_v1	expl.	60	32	48	100 / 4.5	91 / 1.5
av2_cpu_v2	expl.	30	38	58	100 / ∞	95 / 1.7
av2_cpu_v2	expl.	40	34	54	100 / ∞	92 / 1.7
av2_cpu_v2	expl.	60	39	55	100 / 4.9	87 / 1.5

agent-level certify-or-recalibrate rule covers $\geq 90\%$ of held-out scenes in only 46–54% of draws (Table V): its guarantee is not merely loose but wrong. Scene-level Learn-then-Test (Sec. III-E) restores it. With the Hoeffding–Bentkus p -value it is valid but returns an infinite region for the smallest budgets and a region 4.5–4.9 $\times$ the oracle’s at $m = 60$ scenes; with the betting p -value it covers $\geq 90\%$ of held-out scenes in 88–92% of draws at $m = 40$ (region 1.6–1.8 $\times$ the oracle’s) and 87–91% at $m = 60$ (1.5–1.7 $\times$), i.e. at the nominal level up to the sampling noise of the small held-out scene pool. Scene-level rates here are *exploratory*; the registered confirmatory test is pending.

J. Comparison with an online method

Adaptive conformal inference [10] needs a stream of labels. Its running coverage stays within 2 points of nominal after a median of 83–100 labels with $\gamma = 0.05$ but only after 342–696 labels with $\gamma = 0.005$. Frozen after 1000 labels, its threshold covers $\geq 90\%$ of held-out scenes in only 40–50% of streams (like plain recalibration): ACI gives long-run, not finite-sample, control, which is what an audit-then-deploy workflow needs.

K. H8: robustness across a model zoo

Table VI is the full forward-direction model-zoo comparison: all 3 of 3 planned models. The under-coverage direction is consistent across all three (gap ≥ 0.02 at $\alpha = 0.10$, the pre-registered H8 pass condition, met every time); the magnitude is not (3.3 to 10.5 points, a $3.2\times$ range), and neither is the effect size of any repair attempted on it (H4’s gap reduction ranges 12–49%; H6’s ρ ranges 0.10–0.39). H8 is therefore supported on the narrow pre-registered clause (gap replicates, sign never flips) and is itself the source of the paper’s main qualification: solution effect sizes measured on one model should not be assumed to transfer to another, even within the same architecture and training recipe family.

L. H9: reverse direction (nuScenes $\rightarrow$ Argoverse 2)

[TODO: Pending: nuScenes-source model training is queued behind the train-split conversion (in progress). Fill in once results/run2/ns_cpu_v1__reverse.json exists.]

VI. DISCUSSION AND LIMITATIONS

A planner that trusts a source-calibrated region on unseen deployment data is, on this evidence, trusting a region that is too small: at $\alpha = 0.10$ the true miscoverage across our three models ranges from $\alpha + 0.03$ to $\alpha + 0.11$, not α . That gap is silent — nothing in the deployed system signals it — which is precisely why we treat measurement and audit as first-class contributions and not a prelude to the “real” fix. Three results bear directly on practice. First, covariate reweighting is not a safe default here: it made the gap worse on *all three* models tested (Sec. V-B), so a practitioner who reweights and stops is worse off than one who does nothing. Second, the label-free normalised score (H4) is not a substitute for labels either — across the three models it repaired 49%, 12%, and 31% of the gap, and nothing observed at deployment time (without target labels) currently tells you which regime you are in. Third, the coverage audit (H5b) is the one method that behaved as pre-registered on *all three* models without exception: with on the order of 10^3 labelled target scenes it reliably detects a violation of this size at a low false-alarm rate, on every model we tried it on. The practical recommendation this supports is modest but concrete: budget for a few hundred to one thousand labelled target-domain scenes before trusting a transferred conformal region, and treat covariate reweighting and the normalised score as diagnostics to try, not guarantees to rely on.

a) *Why the effect size varies by model.*: The three models (Table VI) differ in training data size and schedule, and their gaps span a $3.2\times$ range (3.3 to 10.5 points) with no monotonic relationship to point-prediction accuracy — the most accurate of the three (av2_cpu_v2, lowest minADE₆) has the *largest* coverage gap. We do not have a causal explanation for this from three data points; it is exactly the kind of pattern a still-larger model zoo would be needed to explain, and we report it as an open question rather than force an explanation onto $n = 3$.

b) *Limitations (fixed).*: (i) Two datasets and one architecture family; (ii) CPU-trained models that are quantifiably below published state-of-the-art point accuracy on this benchmark (+28% to +59% minADE₆ versus a properly-resourced baseline, Sec. IV) — our pre-registered 15% tolerance for this was exceeded, so solution effect sizes measured here should not be assumed to transfer to a stronger model without further study; (iii) the observed gap is an association—its cause (annotation, sampling, agent mix) is not isolated, only a sampling-rate confound was tested; (iv) marginal, not conditional, coverage; (v) hypotheses added after Run 1 are labelled post-hoc; (vi) Proposition 1 covers scale shifts only.

REPRODUCIBILITY

All code, configuration files, pre-registered hypotheses, and the append-only experiment log are in the project repository (paper01-coverage-transfer). Every number and table in this paper is regenerated from the raw result files by paper/make_results.py; no result is typed by hand.

DATA AVAILABILITY

Argoverse 2 [25] and nuScenes [26] are third-party, openly available research datasets (Argoverse 2: <https://www.argoverse.org/av2.html>; nuScenes: <https://www.nuscenes.org>). **[TODO: Add code/results repository URL and archive DOI once the project is made public (see title_page.pdf for the format used on a companion paper: GitHub link + a Zenodo archive DOI under CC BY 4.0).]**

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work the corresponding author (H. W. Saleem) used Anthropic Claude to edit language, draft and revise text, write and debug analysis code, run and monitor experiments, and generate figures and tables from computed results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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TABLE VI

COVERAGE TRANSFER BY MODEL AND DIRECTION, $\alpha = 0.10$. “RED.” = % REDUCTION IN GAP FROM THE LABEL-FREE NORMALISED SCORE (H4); “POWER/FA” = AUDIT REJECT PROBABILITY AT $k = 1000$ TARGET LABELS ON THE SHIFTED / SAME-DOMAIN SPLIT (H5B); ρ = SPEARMAN CORRELATION BETWEEN THE LABEL-FREE DOMAIN-MONITOR AUC AND THE PER-CITY MISCOVERAGE (H6).

Model	Dir.	n_{cal}	n_{tgt}	gap (pt)	95% CI	norm. gap	red. (%)	power/FA @1000	ρ
av2_cpu_v1	forward	4576	9041	3.3	[2.1, 4.5]	1.7	49	0.95/2.4	0.37
av2_valsplit_v1	forward	4576	9041	8.0	[6.4, 9.4]	7.0	12	1.00/1.0	0.10
av2_cpu_v2	forward	4576	9041	10.5	[9.4, 12.2]	7.3	31	1.00/6.2	0.39

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